

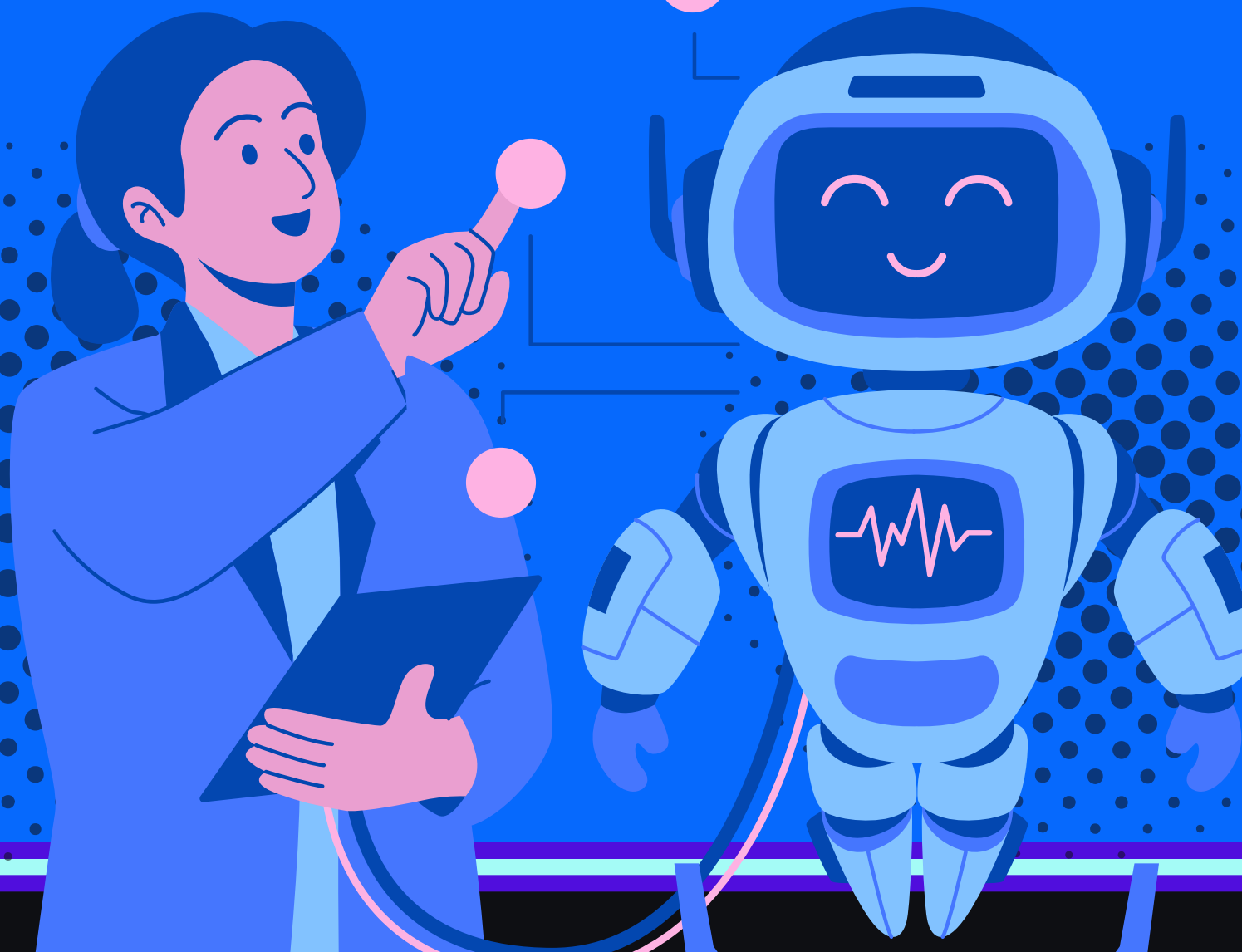


by Kunal Das

AI XCHANGE

AI ENABLER

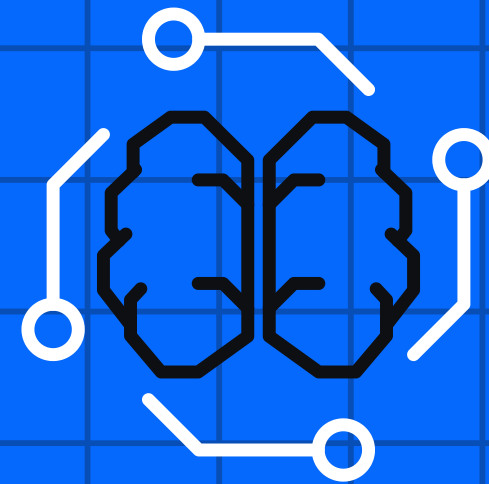
INTELLIGENT LLM ROUTING AT SCALE



DATE November 8, 2025

LOCATION Bangalore

Hello



KUNAL DAS

Developer Advocate APAC, CAST AI

Organizer of
CNCF Kolkata,
Cloud Computing Circle,
Hashicorp User Group Bangalore

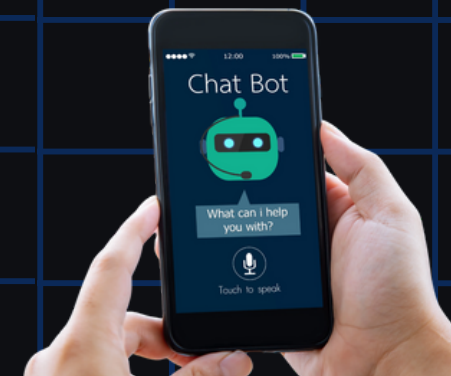
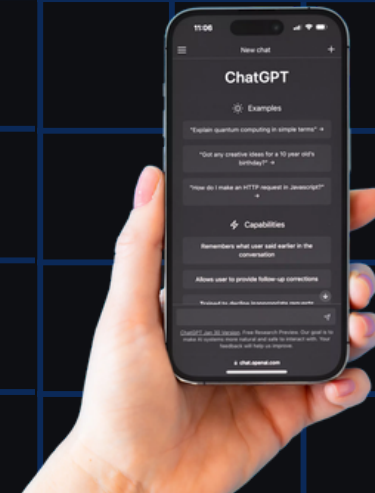
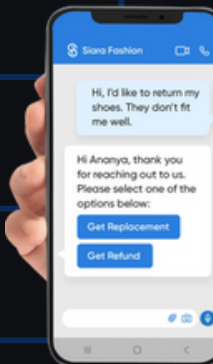
7x Azure, 1x Hashicorp Certified, FinOps
Certified Engineer

How we use LLMs

End users



UI/UX



API

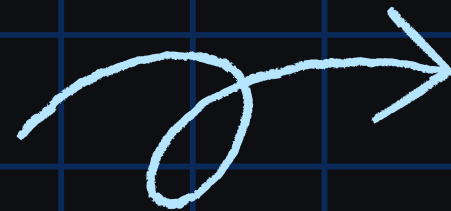


LET'S CHECK SOME PROMPTS

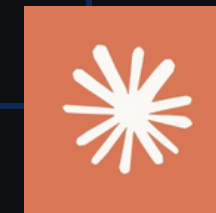


PROMPT1

What is Kubernetes and what are its main components?



GPT-3.5-turbo



Claude 3 Haiku

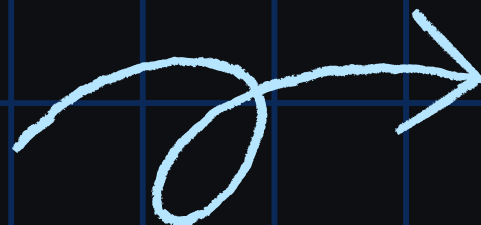


Llama 3.1:8b

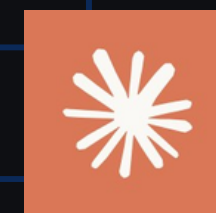
PROMPT2

Analyze this Kubernetes cluster's resource utilization data and provide 3 key optimization recommendations:

- 40% CPU usage across 50 nodes
- 60% memory utilization
- 15 pods in pending state
- Average pod startup time: 45 seconds



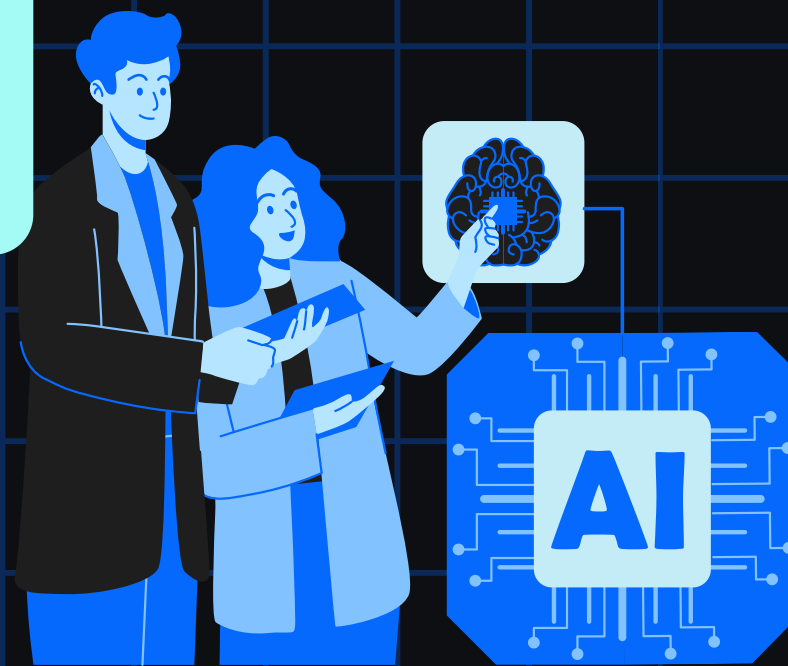
GPT-4o-mini



Claude 3 Sonnet



Llama 3.1:70b



LET'S CHECK SOME PROMPTS

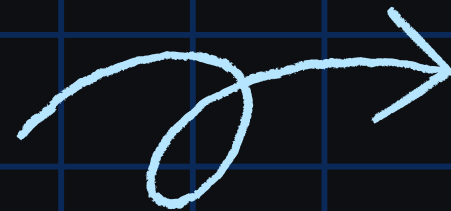


PROMPT 3

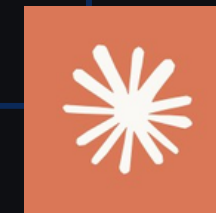
Design a multi-region Kubernetes disaster recovery strategy for a financial services application that:

- *Must maintain 99.99% uptime*
- *Handles sensitive PII data*
- *Requires <100ms latency for transactions*
- *Operates across AWS and GCP*
- *Has compliance requirements for GDPR and SOC2*

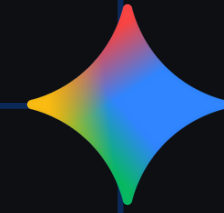
Provide architecture decisions, security considerations, cost implications, and implementation phases.



GPT-4o



Claude 3.5 Sonnet

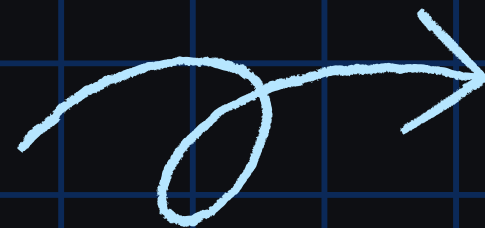


Gemini 1.5 Pro

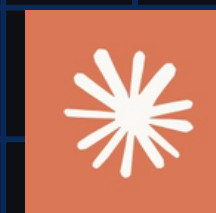
PROMPT 4

Create a Python script that:

- *Connects to a Kubernetes cluster using the official client*
- *Identifies pods with CPU usage > 80%*
- *Automatically scales their deployments*
- *Sends Slack notifications with scaling actions*
- *Includes error handling and logging 6. Add unit tests for the scaling logic*



GPT-5



Claude 4 Opus



Deepseek-R1:14b



PURPOSE DRIVEN LLM

Search & Fact checking

These models excel at grounding responses in current data and cross-referencing multiple sources for truth verification.



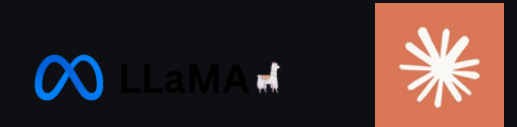
Deep research thinking

These models handle intricate queries that demand deep understanding, context retention, and nuanced decision-making.



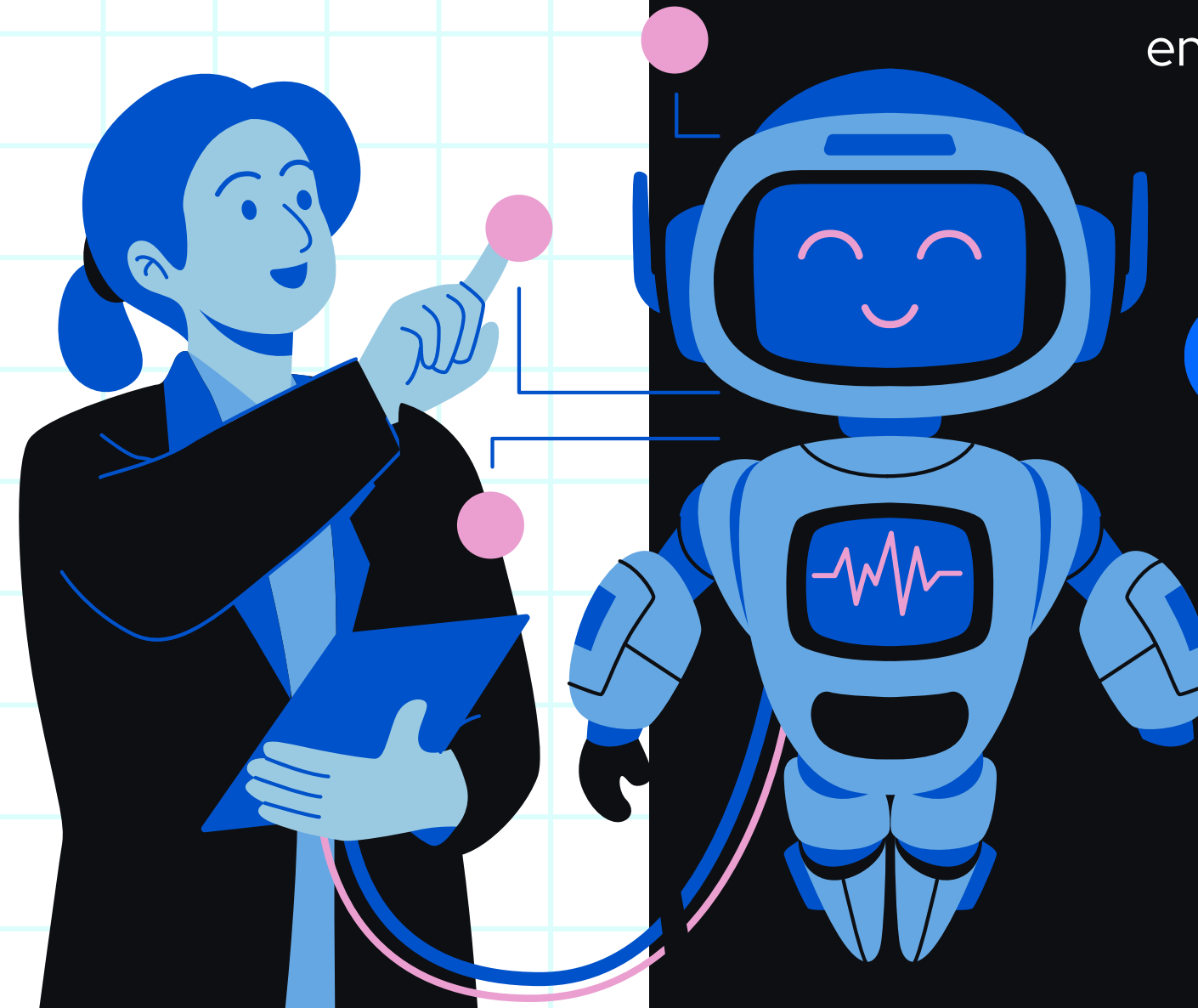
Integrations

These lightweight models provide compatible endpoints across all providers, ensuring smooth workflow automation.

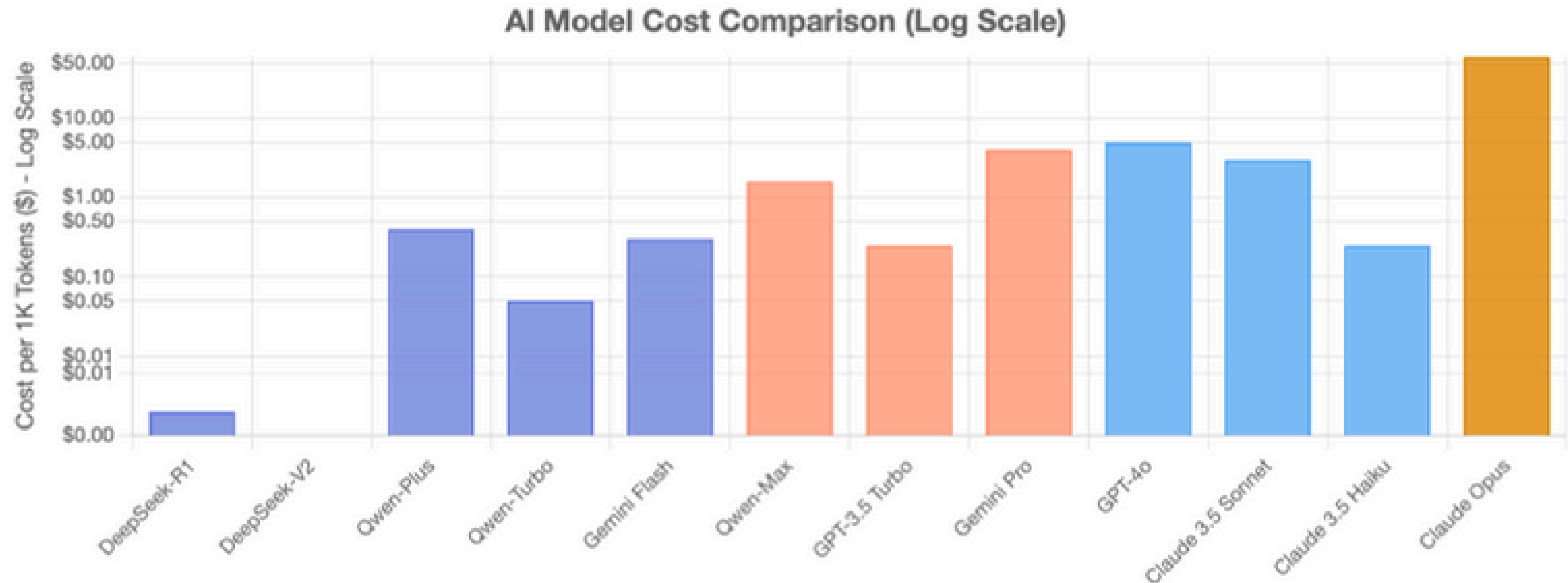
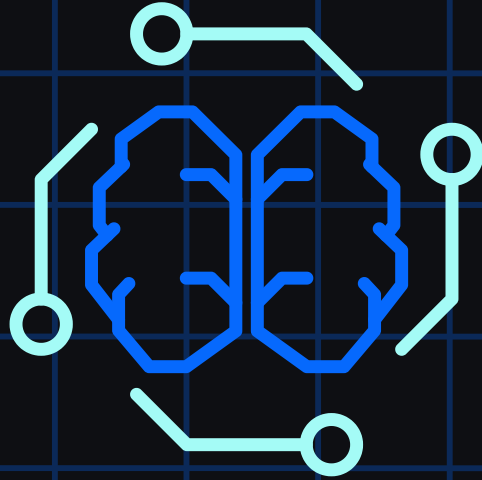


Coding / Development

These models understand code context, suggest optimizations, and can handle complex refactoring operations.



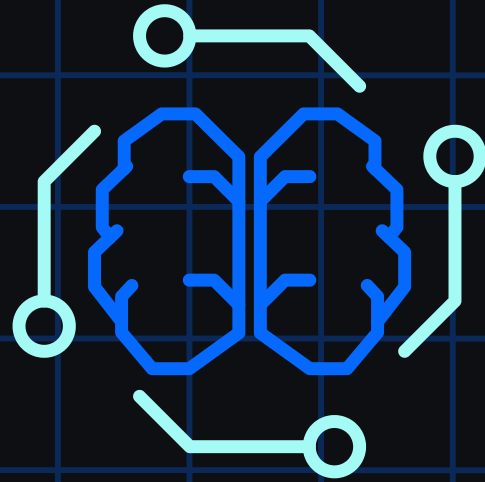
Let's Talk About Money



⚠️ **CRITICAL INSIGHT:** Using Claude 3 Opus (\$60/1K) vs DeepSeek-V2 (\$0.001/1K) = 60,000x cost difference!



Let's Talk About Money



💎 **Budget Champions**
< \$0.50/1K tokens
DeepSeek, Gemini Flash, Qwen, Alibaba Models

⚡ **Balanced Performance**
\$0.50 - \$5.00/1K tokens
GPT-4o, Gemini Pro, OpenAI GPT-3.5

🎯 **Premium Quality**
\$5.00 - \$15.00/1K tokens
Claude 3.5 Sonnet, GPT-4o Mini

🏆 **Enterprise Grade**
> \$15.00/1K tokens
Claude 3 Opus, Claude 3.5 Haiku



LLM HOSTING

Self-Hosting in VM

- Deploy models directly on virtual machines
- Best for: Low-volume workloads, testing, single-application use cases

Kubernetes Deployment

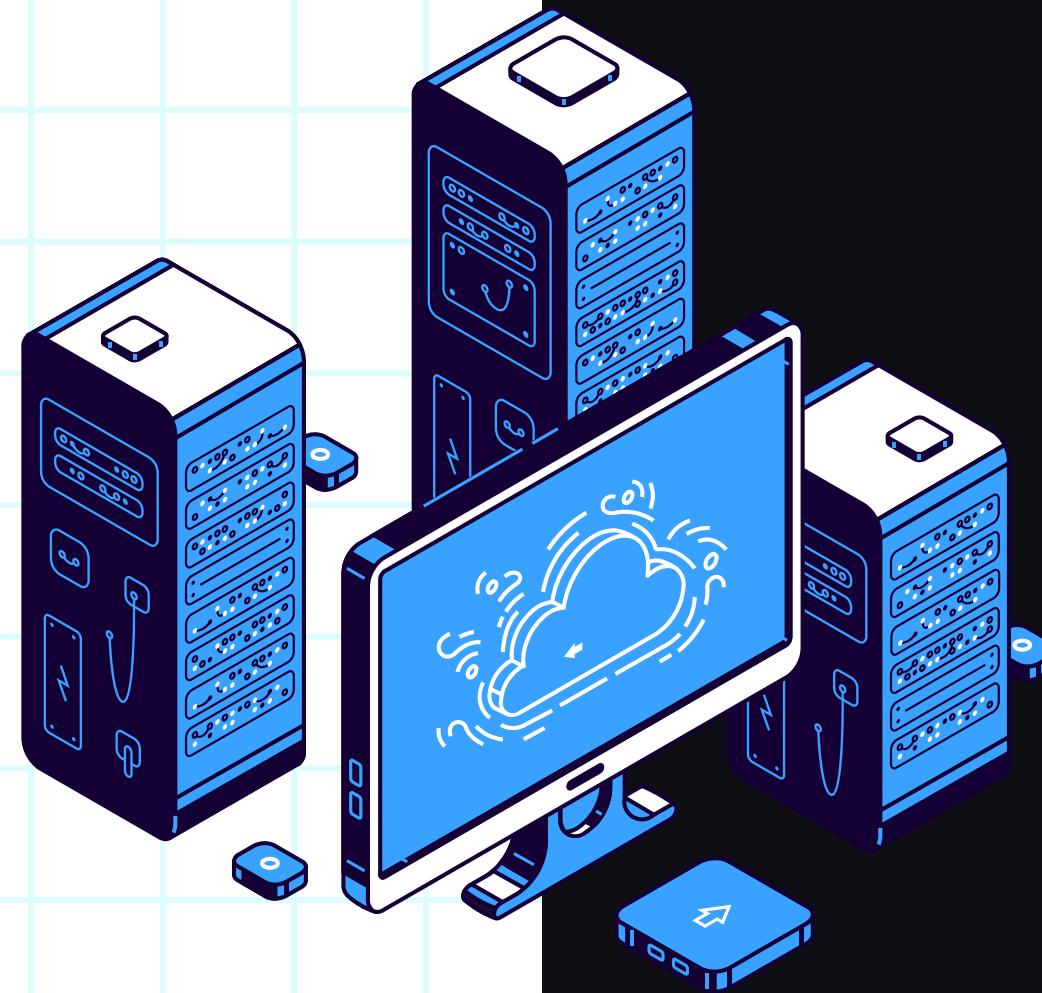
- Auto-scaling LLMs with in your cluster
- Best for: High availability, multi-tenant apps, microservices architecture

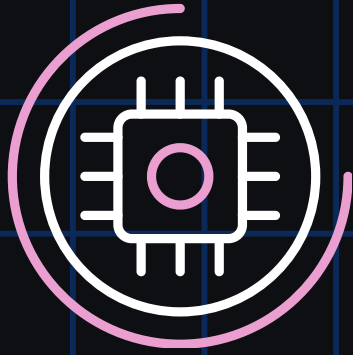
API-Based

- write logic to call different APIs as per requirements
- Best for: Less setup time, rapid iteration

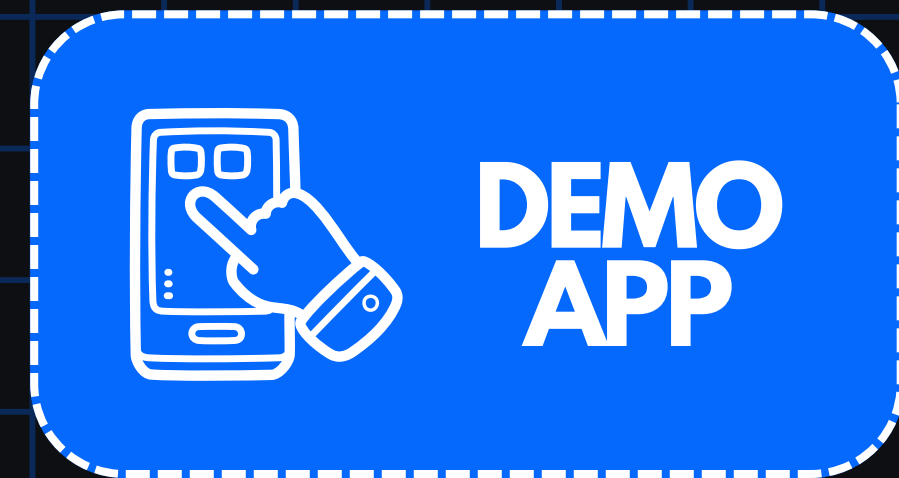
Fallback Mechanism

- Automatic failover when primary models fail or rate-limit
- Best for: Production systems, SLA requirements, business continuity

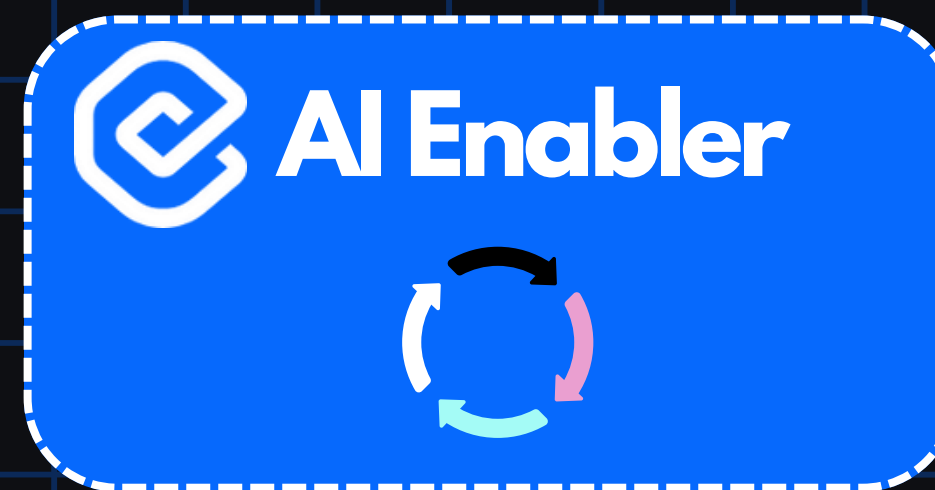
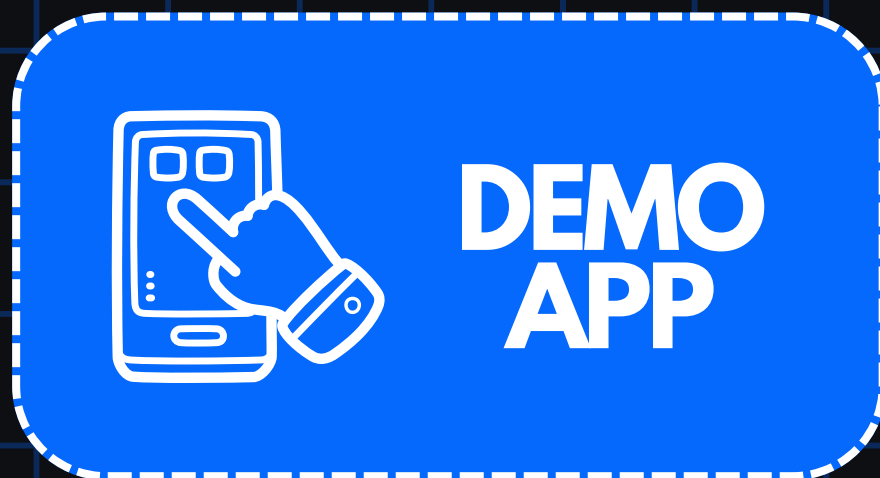




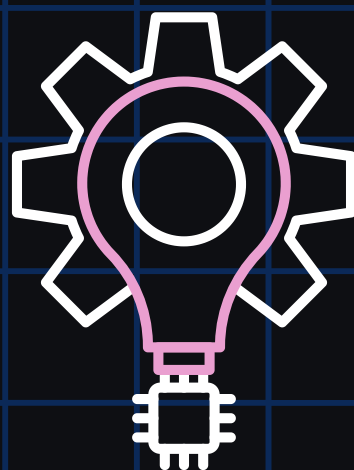
AI ENABLER

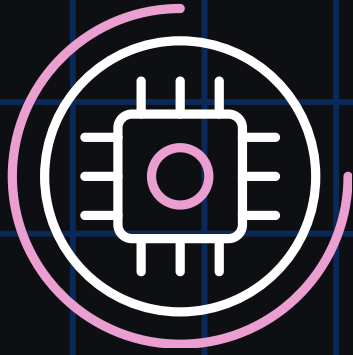


Open AI models

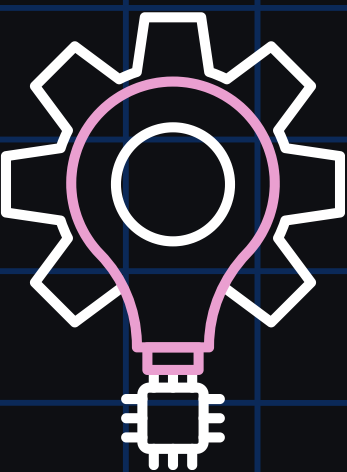
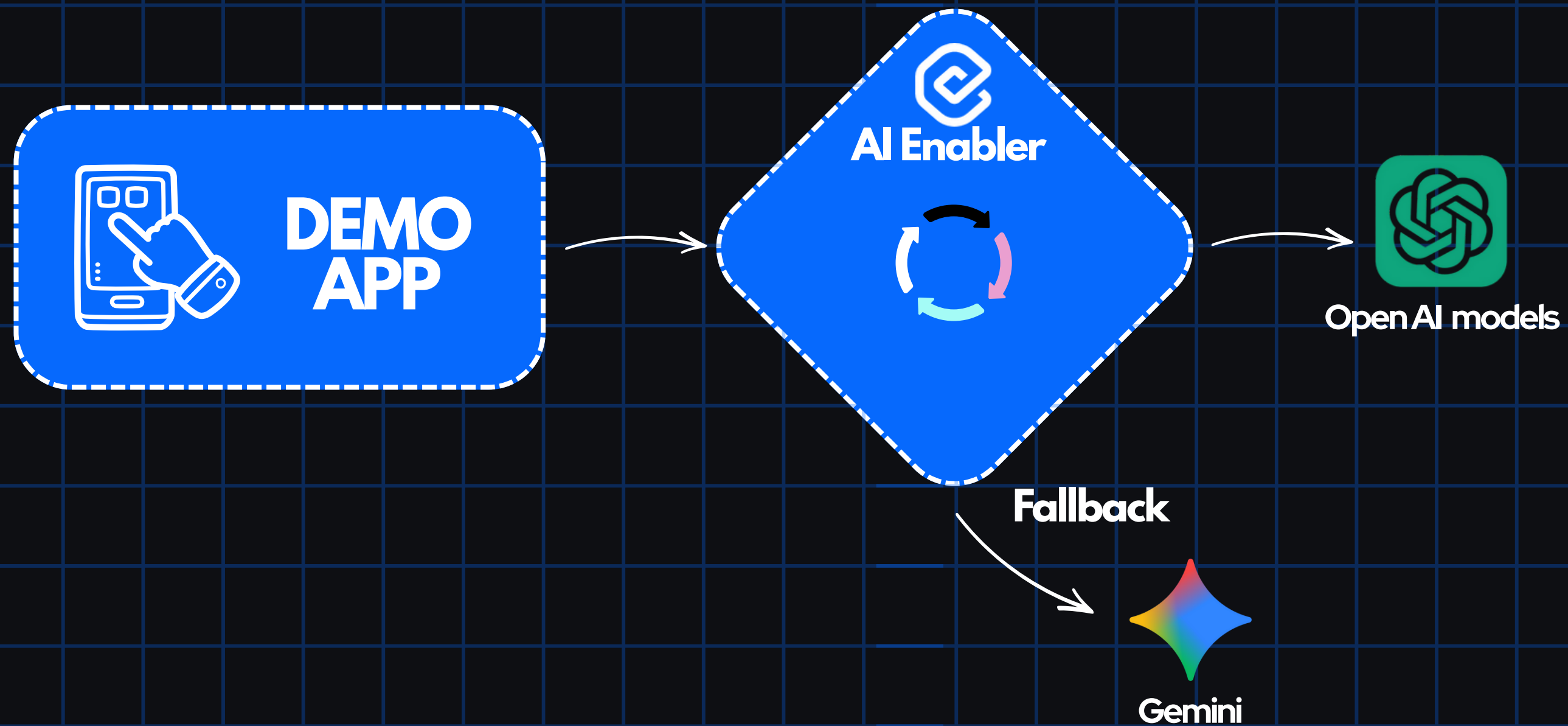


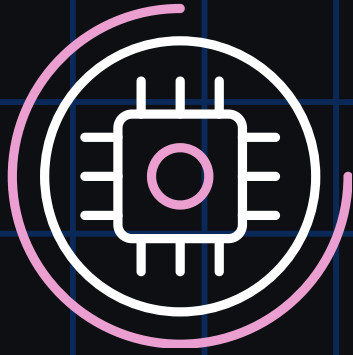
Open AI models



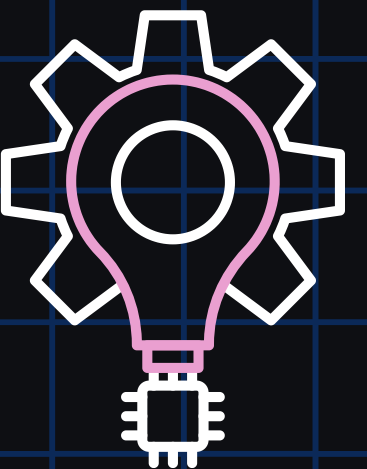
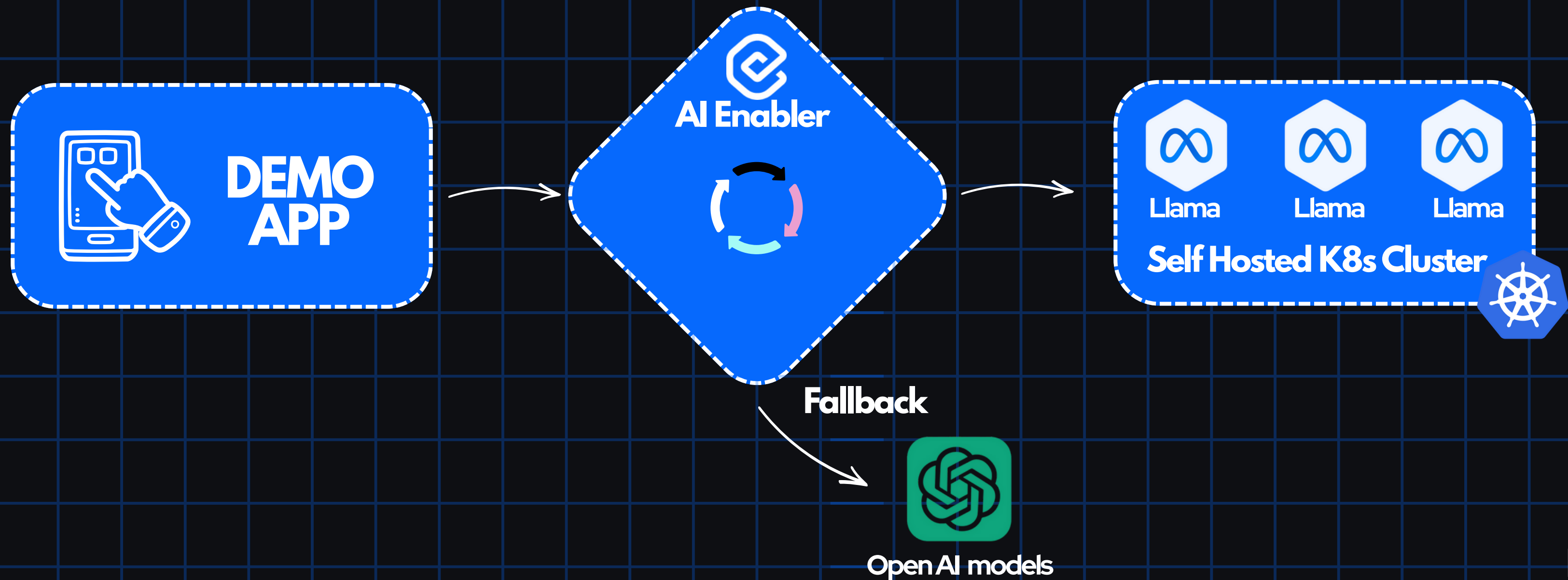


AI ENABLER





AI ENABLER

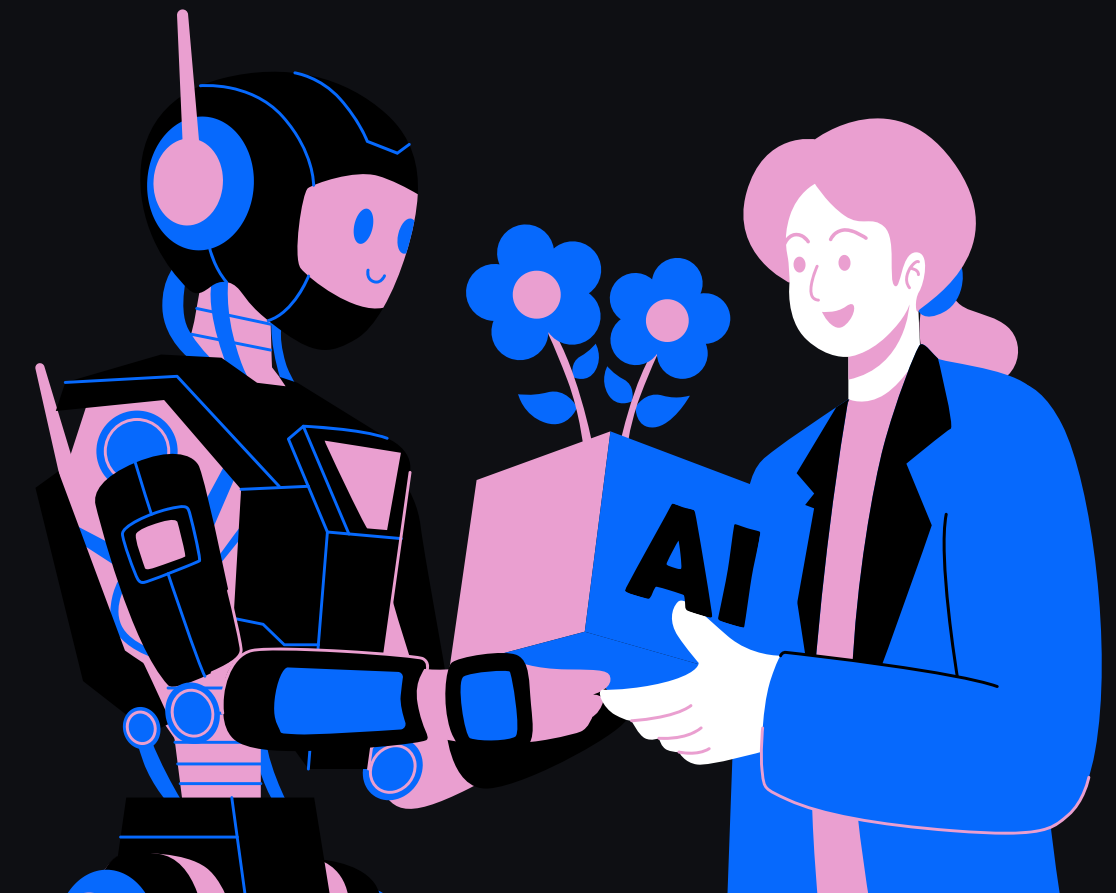


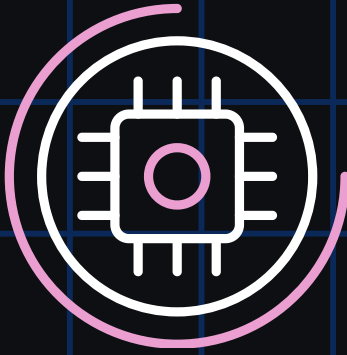
WHAT WE ACHIEVED

Multi-Provider Freedom

Enterprise-Ready Reliability

Cost Savings





RELAIAABILITY??

Cybernews

Thousands of users report Amazon issues

1 day ago



Reuters

Amazon says AWS cloud service back to normal after outage disrupts businesses worldwide

Disruptions to its servers can cause outages across websites and platforms - ranging from food delivery apps to gaming platforms and airline...

2 weeks ago



CRN Magazine

Outflare Blames Google Cloud For Mass Services Outages

Outflare blamed an outage at Google Cloud for a number of products and services by content delivery network vendor going down Thursday.

Jun **Mint**

Microsoft Azure down: What caused the outage? Which services are affected?

1 hour ago



Zoom Bangla News

Microsoft Outage Today: Why Are Azure, 365, and Teams Down Across the U.S.?



Times Now

Microsoft Azure Down: Server Outage Impacts Multiple Services Including 365,...

6 minutes ago



ET The Economic Times

Azure down: Thousands of users complain about outage; here's Microsoft's latest...

2 hours ago



amazon

BREAKING NEWS

AMAZON'S CLOUD SERVICE IS DOWN AGAIN



Mashable

Google Cloud down: The GCP outages explained

Google reported that Google Cloud Platform (GCP), its popular cloud hosting service, experienced a problem Thursday afternoon.

12 Jun 2025

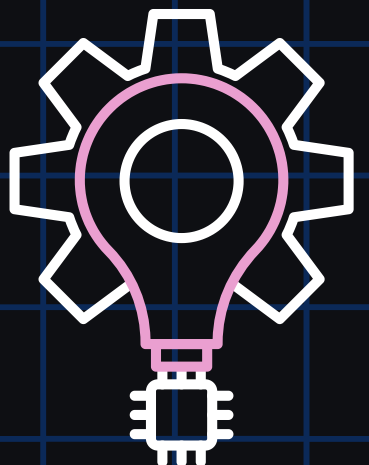


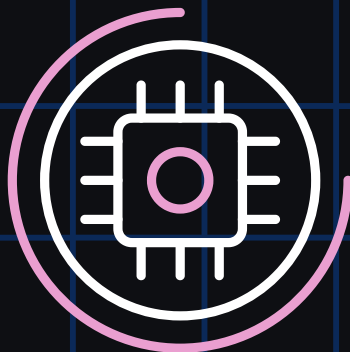
WIRED

Security News This Week: Amazon Explains How Its AWS Outage Took Down the Web

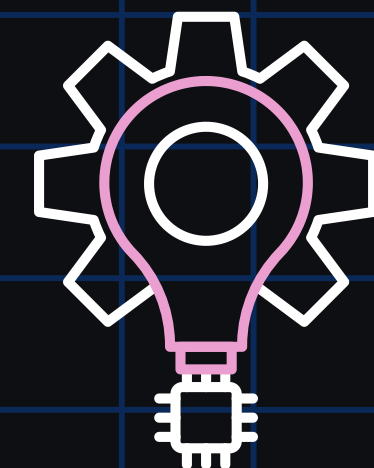
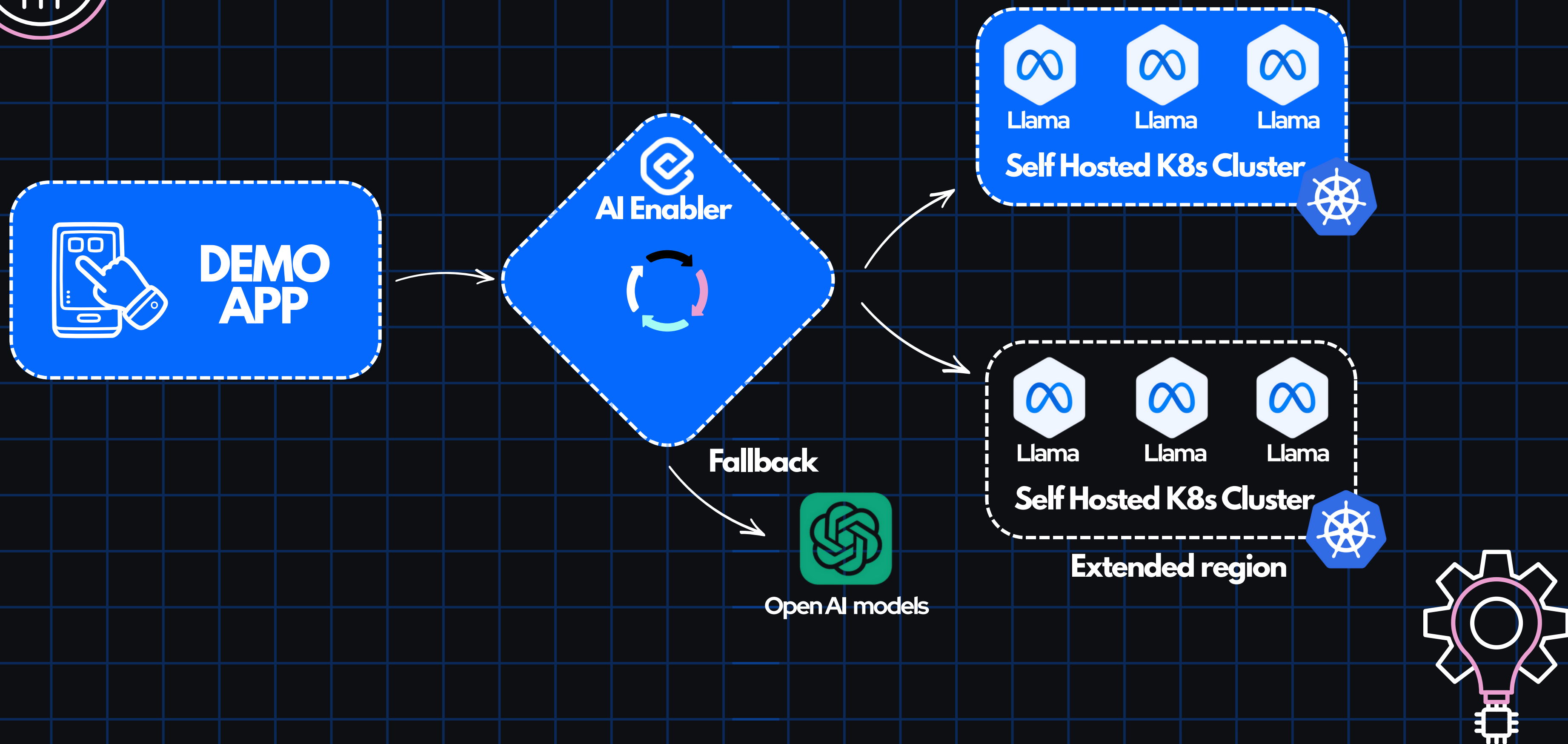
AWS confirmed in a "post-event summary" on Thursday that its major outage on Monday was caused by Domain System Registry failures in its...

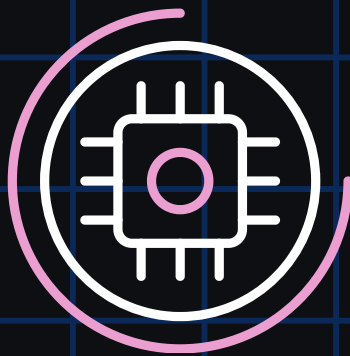
2 weeks ago



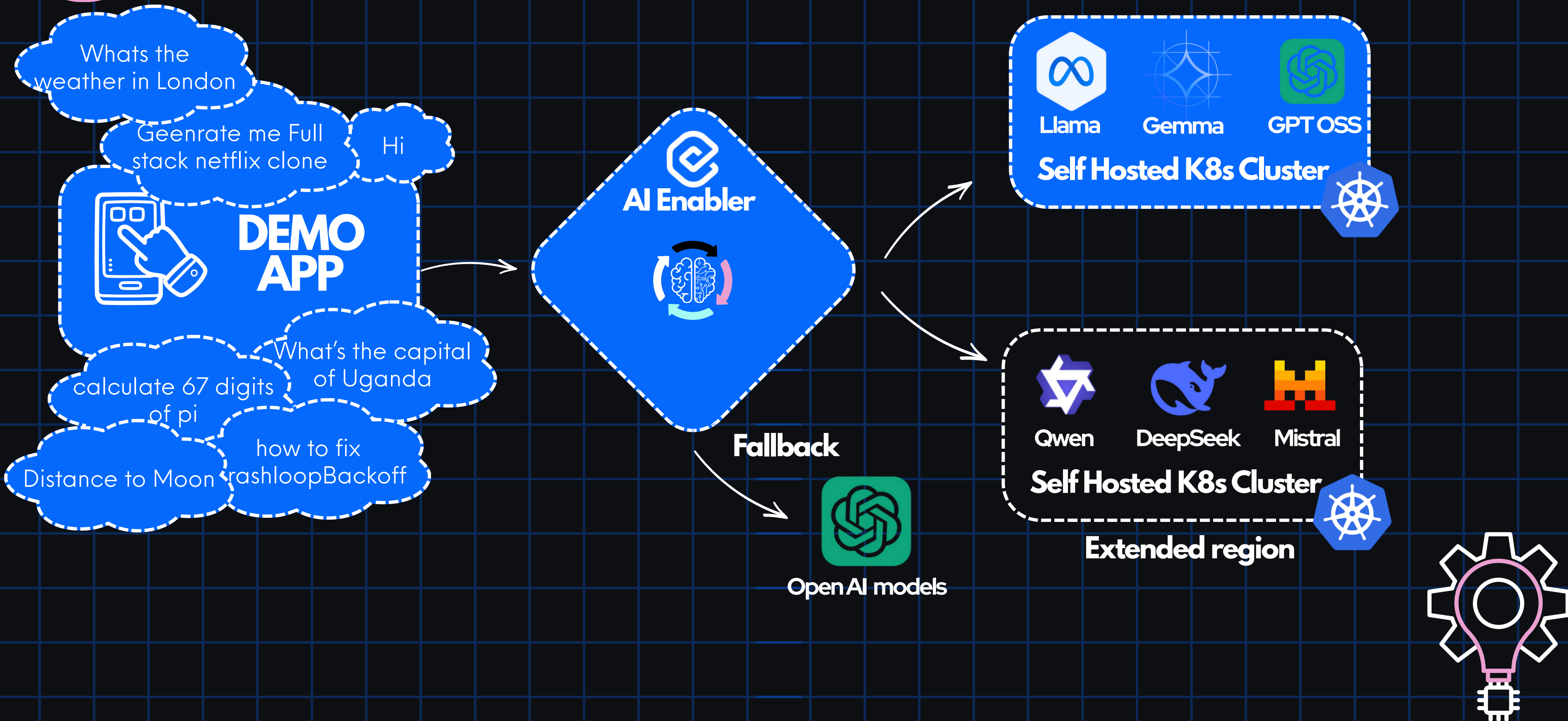


AI ENABLER





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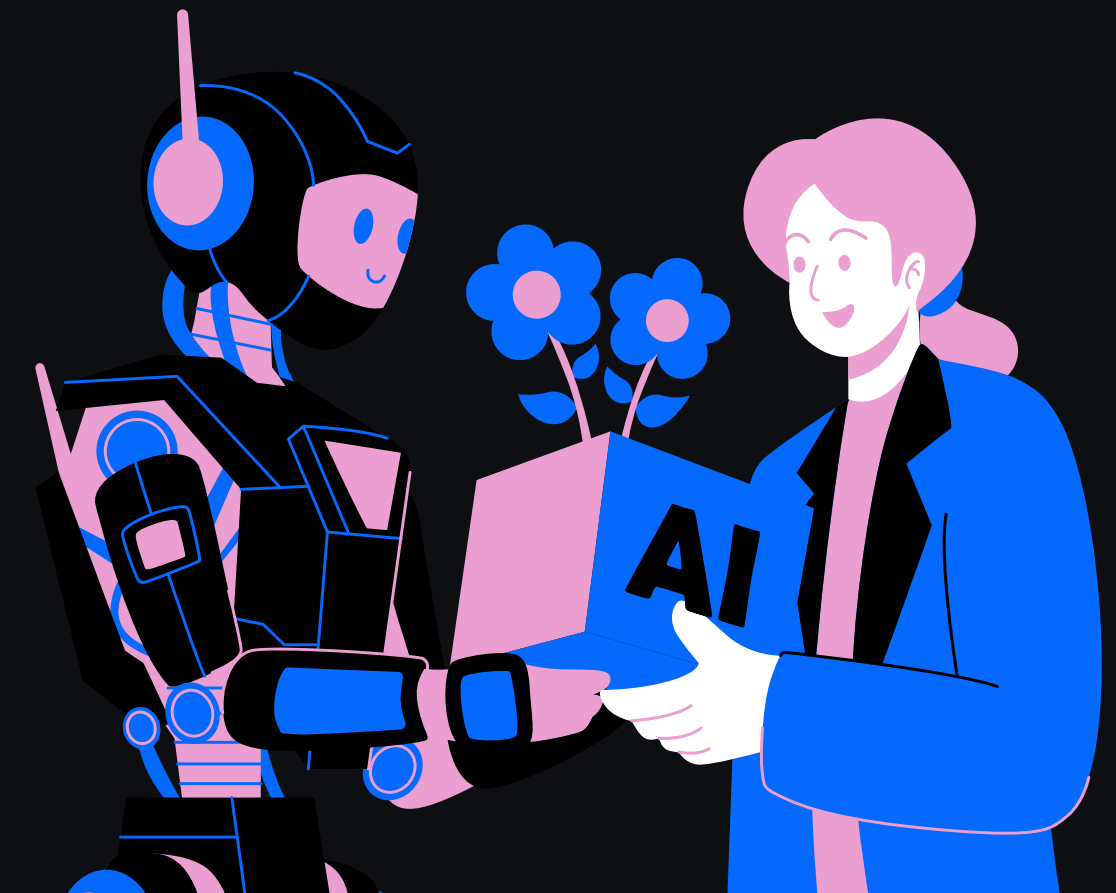
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Enterprise-Ready Reliability

Intelligent Cost Savings

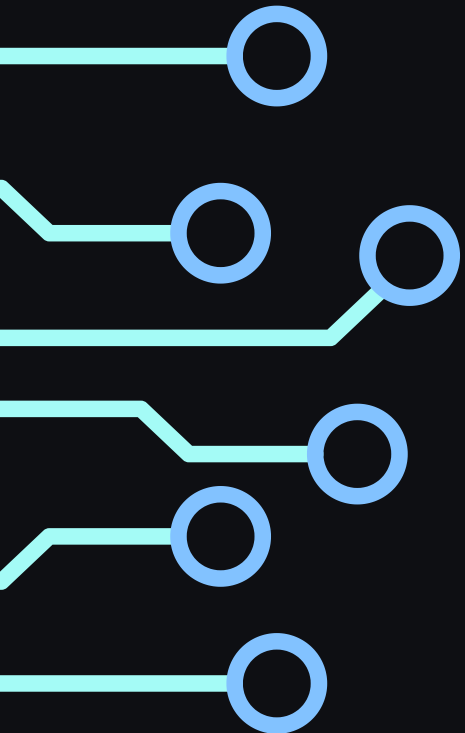
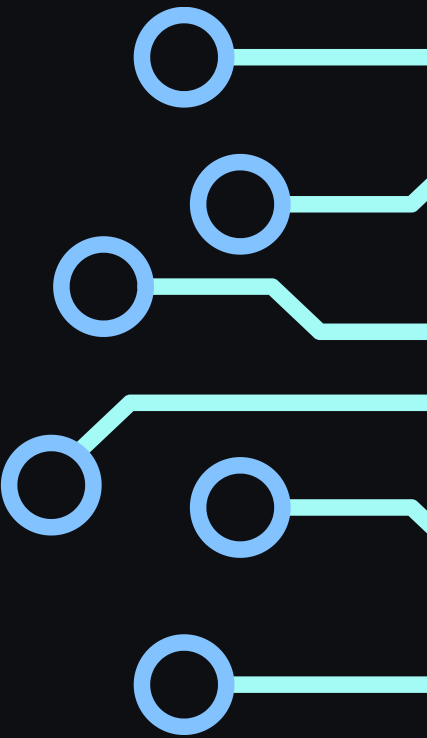
Quality-First Routing



CONCLUSION

- ✓ Intelligent routing saves money
- ✓ Quality doesn't have to cost premium prices
- ✓ Open-source Llama competes with proprietary models
- ✓ CAST AI makes it simple to implement

“The best LLM isn't the most expensive one - it's the RIGHT one for each task. ”



THANK YOU

For your attention

PING ME
ANYTIME TO
KNOW MORE

